

MIT OpenCourseWare

5.60

Thermodynamics and Kinetics 5.60 — Thermodynamics and Kinetics May 2026

EXTREMELY HARD – MIT LEVEL**Time Limit: 3 hours****Total Marks: 100**

1. (4 points) Analyze the limitations of standard approaches to Statistical Thermodynamics when applied to edge cases in 5.60. Construct explicit counterexamples showing where intuitive reasoning fails.
2. (3 points) Analyze the limitations of standard approaches to Chemical Kinetics when applied to edge cases in 5.60. Construct explicit counterexamples showing where intuitive reasoning fails.
3. (7 points) Analyze the limitations of standard approaches to Reaction Mechanisms when applied to edge cases in 5.60. Construct explicit counterexamples showing where intuitive reasoning fails.
4. (3 points) Prove the fundamental theorem concerning Catalysis in the context of 5.60 (Thermodynamics and Kinetics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
5. (3 points) Analyze the limitations of standard approaches to Electrochemistry when applied to edge cases in 5.60. Construct explicit counterexamples showing where intuitive reasoning fails.
6. (5 points) Analyze the limitations of standard approaches to Statistical Thermodynamics when applied to edge cases in 5.60. Construct explicit counterexamples showing where intuitive reasoning fails.
7. (5 points) Prove the fundamental theorem concerning Chemical Kinetics in the context of 5.60 (Thermodynamics and Kinetics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
8. (4 points) Analyze the limitations of standard approaches to Reaction Mechanisms when applied to edge cases in 5.60. Construct explicit counterexamples showing where intuitive reasoning fails.

9. (6 points) Prove the fundamental theorem concerning Catalysis in the context of 5.60 (Thermodynamics and Kinetics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
10. (4 points) Analyze the limitations of standard approaches to Electrochemistry when applied to edge cases in 5.60. Construct explicit counterexamples showing where intuitive reasoning fails.
11. (5 points) Prove the fundamental theorem concerning Statistical Thermodynamics in the context of 5.60 (Thermodynamics and Kinetics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
12. (4 points) Construct an original proof-level problem involving Chemical Kinetics for 5.60 (Thermodynamics and Kinetics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
13. (4 points) Construct an original proof-level problem involving Reaction Mechanisms for 5.60 (Thermodynamics and Kinetics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
14. (3 points) Construct an original proof-level problem involving Catalysis for 5.60 (Thermodynamics and Kinetics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
15. (2 points) Prove the fundamental theorem concerning Electrochemistry in the context of 5.60 (Thermodynamics and Kinetics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
16. (4 points) Analyze the limitations of standard approaches to Statistical Thermodynamics when applied to edge cases in 5.60. Construct explicit counterexamples showing where intuitive reasoning fails.
17. (2 points) Construct an original proof-level problem involving Chemical Kinetics for 5.60 (Thermodynamics and Kinetics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
18. (2 points) Prove the fundamental theorem concerning Reaction Mechanisms in the context of 5.60 (Thermodynamics and Kinetics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
19. (4 points) Analyze the limitations of standard approaches to Catalysis when applied to edge cases in 5.60. Construct explicit counterexamples showing where intuitive reasoning fails.

20. (3 points) Analyze the limitations of standard approaches to Electrochemistry when applied to edge cases in 5.60. Construct explicit counterexamples showing where intuitive reasoning fails.
21. (5 points) Analyze the limitations of standard approaches to Statistical Thermodynamics when applied to edge cases in 5.60. Construct explicit counterexamples showing where intuitive reasoning fails.
22. (4 points) Prove the fundamental theorem concerning Chemical Kinetics in the context of 5.60 (Thermodynamics and Kinetics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
23. (4 points) Construct an original proof-level problem involving Reaction Mechanisms for 5.60 (Thermodynamics and Kinetics) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
24. (4 points) Prove the fundamental theorem concerning Catalysis in the context of 5.60 (Thermodynamics and Kinetics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
25. (6 points) Prove the fundamental theorem concerning Electrochemistry in the context of 5.60 (Thermodynamics and Kinetics). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

Solutions

Solution 1:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Statistical Thermodynamics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 2:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Chemical Kinetics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 3:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Reaction Mechanisms. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 4:

The proof follows from the definitions and properties established in 5.60. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 5:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Electrochemistry. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 6:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Statistical Thermodynamics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 7:

The proof follows from the definitions and properties established in 5.60. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 8:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Reaction Mechanisms. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 9:

The proof follows from the definitions and properties established in 5.60. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 10:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Electrochemistry. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 11:

The proof follows from the definitions and properties established in 5.60. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 12:

Solution approach: First recall the definitions and key theorems related to Chemical Kinetics from 5.60. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 13:

Solution approach: First recall the definitions and key theorems related to Reaction Mechanisms from 5.60. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 14:

Solution approach: First recall the definitions and key theorems related to Catalysis from 5.60. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 15:

The proof follows from the definitions and properties established in 5.60. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 16:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Statistical Thermodynamics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 17:

Solution approach: First recall the definitions and key theorems related to Chemical Kinetics from 5.60. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 18:

The proof follows from the definitions and properties established in 5.60. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 19:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Catalysis. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 20:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Electrochemistry. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 21:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Statistical Thermodynamics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 22:

The proof follows from the definitions and properties established in 5.60. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 23:

Solution approach: First recall the definitions and key theorems related to Reaction Mechanisms from 5.60. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 24:

The proof follows from the definitions and properties established in 5.60. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 25:

The proof follows from the definitions and properties established in 5.60. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.